

PAVEL LIASHKOV

Senior Software Engineer | Python Expert | Distributed Systems & Infrastructure

pavel.liashkov@protonmail.com | linkedin.com/in/liashkov | github.com/bigbag | liashkov.site

PROFESSIONAL SUMMARY

Senior Software Engineer with 12+ years shipping production Python systems end-to-end. I build backend services that handle 200M+ records/day, payment platforms reconciling across 50+ providers, and infrastructure on AWS/Kubernetes. Day-to-day: FastAPI, PostgreSQL, Redis, gRPC, Docker, CI/CD. Hired 20+ engineers through 400+ interviews. Active open-source contributor with 2.4M+ PyPI downloads (220K+ monthly). Currently working on multi-agent LLM pipelines and data-intensive microservices.

PROFESSIONAL EXPERIENCE

Backend Engineer | SplitMetrics | Remote

July 2022 – Present

- Wrote a multi-agent research system end-to-end: PydanticAI-based pipeline with 20 skill agents, parallel execution, streaq Redis-backed task queue, and AG-UI SSE streaming with run cancellation
- 33-tool FastMCP server + RAG layer: PostgreSQL pgvector for semantic search, DuckDB VSS with ONNX-optimized embeddings. Query latency under 100ms (INT8 quantization, hybrid HNSW + BM25)
- Ran Snowflake/StarRocks/Airflow data pipelines at 200M+ records/day, 99.9% uptime
- BERT + ONNX keyword analysis service: sub-second responses at 5K+ req/s
- Added offline evaluation pipeline (deterministic checks, LLM judge, groundedness scoring) and a memory extraction service with watermark-based batch processing
- AWS infrastructure (ECS, RDS, S3, Bedrock), Kubernetes, Jenkins CI/CD. Wired multi-provider LLM routing across OpenRouter, DeepSeek, Ollama, and OpenAI-compatible endpoints
- Built enterprise microservices using gRPC, FastAPI, and Celery integrating OpenAI GPT, Claude API, and Apple Search Ads API

Senior Software Engineer | Exness | Limassol, Cyprus

August 2016 – April 2022

- Architected payment processing platform (22+ microservices) with JWS multi-signature validation, state machine invoice lifecycle, and Kafka event-driven reconciliation across 50+ payment providers
- Integrated Bitcoin Core RPC and USDT-OMNI for crypto deposits/withdrawals with wallet management and blockchain confirmation tracking
- 400+ technical interviews conducted; hired 20+ engineers across backend and platform teams
- Implemented PCI DSS-compliant authentication with HashiCorp Vault credential management and JWS cryptographic signing
- Built KYC verification rules engine integrating 12+ MetaTrader servers with SumSub and multi-database connectivity

Software Engineer | Game Stream Joint LLC | Minsk, Belarus

August 2014 – August 2016

- Built microservice-based coupon system and premium shop redesign using Python/Django, increasing user engagement and sales; developed REST APIs for external platform integration with PostgreSQL and Redis

Software Developer | Mobispot Social Systems

October 2012 – August 2014

- Led development of billing systems for payment terminals and vending machines using Python, Flask, and MySQL

KEY PROJECTS

Open Source Libraries — 2.4M+ total downloads, 220K+ monthly on PyPI

- starlette-request-id (1.4M downloads, 120K/mo): ASGI middleware for distributed tracing with context propagation across async operations
- request-id-helper (970K downloads, 100K/mo): Thread-safe request ID tracking for cross-service tracing in microservices
- Additional: starlette-i18n, sqlalchemy-state-machine, prometheus-pusher

E-Paper Reader Projects (Open Hardware)

- papyrix-reader (★ 380): Open-source firmware for Xteink X4/X3 e-readers on ESP32-C3 (~380KB RAM). Single binary auto-detects device via I²C scan. 6 format parsers, Knuth-Plass line breaking, multi-script rendering (Arabic RTL, Thai, CJK), streaming font system. C++20, PlatformIO, FreeRTOS

- epub-to-xtc-converter (★ 90): Browser-based converter with CEngine WASM, Floyd-Steinberg dithering, 42 languages; papyrix-flasher (★ 20): Cross-platform Go flashing utility with SLIP framing and auto-detection

EDUCATION

Master of Science in Information Technology (MSIT) — 2024

University of the People | GPA: 3.97/4.0

Master's Degree in Theoretical Physics — 2005

Kemerovo State University

CERTIFICATIONS & TRAINING

- CKA: Certified Kubernetes Administrator | Astronomer Certification for Apache Airflow 2 | PSD I (Scrum)
- Kaggle Competitions Expert — top 3% (rank 1,712 highest), 3 bronze medals
- Oxford Machine Learning Summer School (OxML 2024) | Udacity ML Engineer Nanodegree | Yandex Go Developer Professional
- Hugging Face AI Agents Course (2025)
- Coursera: Generative AI for Software Development, Agentic AI for Leaders, Data Structures & Algorithms, Self-Driving Cars

TECHNICAL SKILLS

Languages: Python (12+ years), Go (6+ years), SQL, gRPC, Protocol Buffers, REST, GraphQL

Databases: PostgreSQL (pgvector), ClickHouse, Snowflake, StarRocks, Redis, DuckDB, BigQuery, MySQL

Cloud & Infrastructure: AWS (ECS, RDS, S3, Bedrock), Kubernetes (CKA), Docker, Helm, ArgoCD, Jenkins, GitHub Actions, GitLab CI/CD

Workflow & Messaging: Apache Airflow, Celery, RabbitMQ, Kafka, Redis Streams, streaq

AI/ML Integration: PydanticAI, OpenAI GPT, Anthropic Claude, ONNX Runtime, RAG, FastMCP, MCP protocol

Observability: Prometheus, Grafana, LangSmith, Pydantic Logfire, Sentry, distributed tracing

Security: HashiCorp Vault, PCI DSS, JWS/JWT, RSA/AES hybrid encryption, Bitcoin Core RPC

Data Processing: Pandas, Polars, PyArrow, SQLAlchemy, Alembic

LANGUAGES

English (Professional working proficiency) | Russian (Native)